Models of Panel Data

Henrique Veras

PIMES/UFPE

Intro

Here we will study models that deal with data sets that combine time series and cross sections: **panel data sets**.

The analysis of panel data allows us to learn about the economic processes while accounting for both **heterogeneity** across units of observations and for the dynamic effects that are not visible in cross-secions.

Panel Data Modeling

panel data models are more oriented toward cross-section analyses

- they are wide but typically short.

Thus, in the typical panel, there are a large number of cross-sectional units and only a few periods.

The General Modeling Framework for Analyzing Panel Data

The basic framework is a regression model of the form

$$\begin{aligned} y_{it} =& \mathbf{x}_{it}'\beta + \mathbf{z}_i'\alpha + \varepsilon_{it} \\ =& \mathbf{x}_{it}'\beta + c_i + \varepsilon_{it} \end{aligned}$$

There are K regressors in $\mathbf{x}_{it}$, not including the constant term.

The heterogeneity, or individual effect, is $\mathbf{z}_i'\alpha$ where $\mathbf{z}_i$ contains a constant term and a set of individual or group-specific variables, which may be observed, such as race, sex, location, and so on; or unobserved, such as family specific characteristics, individual heterogeneity in skill or preferences, and so on, all of which are taken to be constant over time t.

Notice that if all variables in $\mathbf{z}$ are observable, we could estimate the model above by OLS.

The complication, however, arises when c_i is **unobserved**.

Our main objective here will be consistent and efficient estimation of the **partial effects**

$$\beta = \partial E[y_{it}|\mathbf{x}_{it}]/\partial \mathbf{x}_{it}$$

Whether this is possible depends on the assumptions about the unobserved (latent) effects.

The General Modeling Framework for Analyzing Panel Data

We can begin with a **strict exogeneity** assumption for the independent variables,

$$E[\varepsilon_{it}|\mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, c_i] = E[\varepsilon_{it}|\mathbf{X}_i, c_i] = 0$$

This implies that the current disturbance is uncorrelated with the independent variables in every period, past, present, and future.

This assumption is sometimes unnecessarily strong but it is a convenient starting point and useful in some applications.

The General Modeling Framework for Analyzing Panel Data

A looser assumption of contemporaneous exogeneity is sometimes useful:

$$E[\varepsilon_{it}|\mathbf{x}_{it}, c_i] = 0$$

The regression model with this assumption restricts influences of $\mathbf{x}$ on $E[\mathbf{y}|\mathbf{x}, c]$ to the current period.

The General Modeling Framework for Analyzing Panel Data

The crucial aspect of the model concerns the heterogeneity.

A conventional assumption is the **mean independence**,

$$E[c_i|\mathbf{x}_{i1}, \mathbf{x}_{i2}, \ldots,] = \alpha$$

This assumption, which underlines the *random effects* model that we will discuss later, states that the unobserved variables are uncorrelated with the included variables.

This is also a strong assumption and likely to be violated in many empirical applications.

A weaker assumption would be that $E[c_i|\mathbf{x}_{i1}, \mathbf{x}_{i2}, \ldots,] = h(\mathbf{x}_{i1}, \mathbf{x}_{i2}, \ldots,) = h(\mathbf{X}_i)$ for some unspecified but nonconstant function of $\mathbf{X}_i$.

Model Structures

Panel data models can broadly be arranged as follows:

1. **Pooled regression**: If $\mathbf{z}_i$ contains only a constant term, then the OLS estimator provides consistent and efficient estimates of the common α and slope vector β.
2. **Fixed effects**: If $\mathbf{z}_i$ is unobserved, but correlated with $\mathbf{x}_{it}$, then the OLS estimator of β is biased and inconsistent (why?). However, the model

$$y_{it} = \mathbf{x}_{it}'\beta + \alpha_i + \varepsilon_{it},$$

where $\alpha_i = \mathbf{z}_i'\alpha$ embodies all the observable effects and specifies an estimable conditional mean.

Model Structures

Panel data models can broadly be arranged as follows:

3. **Random effects**: If the observed individual heterogeneity is uncorrelated with $\mathbf{x}_{it}$, the model may be formulated as

$$
\begin{aligned}
y_{it} =& \mathbf{x}_{it}'\beta + E[\mathbf{z}_i'\alpha] + \mathbf{z}_i'\alpha - E[\mathbf{z}_i'\alpha] + \varepsilon_{it} \\
=& \mathbf{x}_{it}'\beta + \alpha + u_i + \varepsilon_{it}
\end{aligned}
$$

that is, a linear regression model with a compound disturbance that may consistently, albeit inefficiently, estimated by LS.

Balanced and Unbalanced Panels

As suggested by the preceding discussion, a panel data set consists of n set of observations on individuals.

If each individual is observed the same number of times T, the data set is a **balanced panel**.

An **unbalanced panel** data set is one in which individuals may be observed different number of times.

- In empirical applications, this might be due to **attrition.

A **fixed panel** is one in which the same set of individuals is observed for the duration of the study.

A **rotating panel** is one in which the cast of individuals changes from one period to the next.

Well-Behaved Panel Data

Throughout the analysis we consider assumptions A.1. through A.4., as well as the asymptotic conditions on the matrices $\mathbf{X}'\mathbf{X}/n$ and $(\mathbf{X}'\varepsilon)/\mathbf{n}$.

Exceptions to the assumptions are likely to arise in a panel data set.

- For example, the $\mathbf{x}_i$'s may be correlated across observations as they might be the same individual observed in multiple periods.

The panel data set could be treated as follows:

- Since the total number of rows in $\mathbf{X}$ is now $N = n \times T$, the matrix $\bar{\mathbf{Q}}_n$ is

$$\bar{\mathbf{Q}}_n = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{T}\mathbf{X_i}'\mathbf{X_i} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{Q}_i$$

We then view the set of observations on the ith unit as if they were a single observation and apply our convergence arguments to the number of units increasing without bound.

The Pooled Regression Model

The simplest version of the model assumes

$$y_{it} = \alpha + \mathbf{x}_{it}'\beta + \varepsilon_{it}, i = 1, \ldots, n; t = 1, \ldots, T_i$$

$$E[\varepsilon_{it}|\mathbf{x}_{i1}, \mathbf{x}_{i2}, \ldots, \mathbf{x}_{iT_i}] = 0$$

$$E[\varepsilon_{it}\varepsilon_{js}|\mathbf{x}_{i1}, \mathbf{x}_{i2}, \ldots, \mathbf{x}_{iT_i}] = \sigma_\varepsilon^2 \text{ if } i = j \text{ and } t = s \text{ and } = 0 \text{ if } i \neq j \text{ or } t \neq s$$

Least Squares Estimation of the Pooled Model

In the form above,if the remaining assumptions of the classical model are met, then no further analysis is needed.

OLS is the efficient estimator and inference can reliably proceed along the lines of the methods developed earlier in the course.

Robust Covariance Matrix

Suppose we stack the T_i observations for individual i in a single equation

$$\mathbf{y}_i = \mathbf{x}_i\beta + \mathbf{w}_i$$

In a panel data set, although heteroskedasticity might be present, the more substantive effect is cross-observation correlation (or autocorrelation).

Let's assume that the disturbances vector consists of ε_{it} plus an omitted individual component. Then,

$$\begin{aligned} Var[\mathbf{w}_i|\mathbf{X}_i] =& \sigma_\varepsilon^2 \mathbf{I}_{T_i} + \mathbf{\Sigma}_i \\ =& \mathbf{\Omega}_i \end{aligned}$$

Robust Covariance Matrix

The OLS estimator of β is

$$\begin{aligned}\mathbf{b} =& (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\ =& \left[\sum_{i=1}^{n}\mathbf{X_i'X_i}\right]^{-1}\sum_{i=1}^{n}\mathbf{X}_i'\mathbf{w}_i\end{aligned}$$

Consistency of $\mathbf{b}$ can be established as before.

The true covariance matrix would take the form we saw for the GLS model

$$Asy.Var[\mathbf{b}] = \frac{1}{n}\text{plim}\ \left[\frac{1}{n}\sum_{i=1}^{n}\mathbf{X_i'X_i}\right]^{-1}\text{plim}\ \left[\sum_{i=1}^{n}X_i'\boldsymbol{\Omega}_i X_i\right]\text{plim}\ \left[\frac{1}{n}\sum_{i=1}^{n}\mathbf{X_i'X_i}\right]^{-1}$$

As before, the center matrix must be estimated.

Robust Covariance Matrix

We can estimate $Asy.Var[\mathbf{b}]$ with

$$Est.Asy.Var[\mathbf{b}] = \frac{1}{n}\text{plim}\ \left[\frac{1}{n}\sum_{i=1}^{n}\mathbf{X_i'X_i}\right]^{-1}\text{plim}\ \left[\sum_{i=1}^{n}X_i'\hat{\mathbf{w}}\hat{\mathbf{w}}_i'X_i\right]\text{plim}\ \left[\frac{1}{n}\sum_{i=1}^{n}\mathbf{X_i'X_i}\right]$$

where $\hat{\mathbf{w}}_i$ is the vector of T_i residuals for individual i.

This construction is analogous to the "cluster"-robust estimator developed earlier.

Bootstrapped Covariance Matrix

Bootstrapping offers another approach to estimating an appropriate covariance matrix for the estimator.

Suppose we have n individuals (groups) over T periods.

To compute the **block bootstrap** estimator, we draw random samples of $N = n$ with *repetition*. Then, we repeat the process R times.

After R repetitions, we compute the empirical variance of the R replicates.

The estimator is

$$Est.Asy.Var[\mathbf{b}] = \frac{1}{R}\sum_{r=1}^{R}(\mathbf{b_r} - \bar{\mathbf{b}})(\mathbf{b_r} - \bar{\mathbf{b}})'$$

Robust Estimation Using Group Means

The pooled regression model can also be estimated using the sample means of the data.

Pre-multiplying the regression model by $(1/T)\mathbf{i}'$, where $\mathbf{i}'$ is a row vector of ones,

$$(1/T)\mathbf{i}'\mathbf{y}_i = (1/T)\mathbf{i}'\mathbf{x_i}\beta + (1/T)\mathbf{i}'\mathbf{w}_i$$

or

$$\bar{y}_i = \bar{\mathbf{x}}_i'\beta + \bar{w}_i$$

Heteroskedastic variances are now $\sigma_i^2 = (1/T^2)\mathbf{i}'\mathbf{\Omega_i}\mathbf{i}$.

Why might one want to use this estimator when the full data set is available?

Robust Estimation Using Group Means

Consider the general model

$$\mathbf{y}_i = \mathbf{x_i}\beta + c_i\mathbf{i} + \mathbf{w}_i,$$

where c_i is the (latent) heterogeneous effect.

If the mean independence assumption, $E[c_i|\mathbf{X}_i] = \alpha$ is not met, then the effect will be transmitted to the group means as well.

Robust Estimation Using Group Means

A common specification is Mundlak's (1978), where we employ the projection of c_i on the group means

$$c_i|\mathbf{X}_i = \bar{\mathbf{x}}_i'\gamma + \nu_i$$

Then,

$$\begin{aligned} y_{it} =& \mathbf{x}_{it}'\beta + c_i + \varepsilon_{it} \\ =& \mathbf{x}_{it}'\beta + \bar{\mathbf{x}}_i'\gamma + [\nu_i + \varepsilon_{it}] \\ =& \mathbf{x}_{it}'\beta + \bar{\mathbf{x}}_i'\gamma + u_{it}, \end{aligned}$$

where, by construction, $Cov[u_{it}, \bar{\mathbf{x}}_i] = 0$.

The implication is that the group means estimator estimates not β, but $\beta + \gamma$.

Averaging the observations in the group collects the entire set of effects, observed and latent, in the group means.

Estimator with First Differences

First differencing is another approach to estimation.

Consider the model

$$y_{it} = c_i + \mathbf{x}'_{it}\beta + \varepsilon_{it}$$

which implies the first differences equation

$$\Delta y_{it} = \Delta c_i + (\Delta \mathbf{x}_{it})'\beta + \Delta \varepsilon_{it}$$

or

$$\begin{aligned} \Delta y_{it} =& (\Delta \mathbf{x}_{it})'\beta + \varepsilon_{it} - \varepsilon_{i,t-1} \\ =& (\Delta \mathbf{x}_{it})'\beta + u_{it} \end{aligned}$$

Estimator with First Differences

The advantage of the first difference approach is that it removes the heterogeneity from the model.

The disadvantage is that it removes all time-invariant variables from the model.

- If we have a panel data set and want to estimate the effects of schooling on wages, for example, we would not be able to employ this strategy, as years of education of workers are usually fixed in time.

Note as well that the differencing procedure trades the cross-observation correlation in c_i for a moving average (MA) disturbances $u_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}$, which is autocorrelated.

However, first differencing might be useful when we have a setting involving two time periods, before and after a treatment.

Estimating Treatment Effects with First Differencing

Consider the model

$$y_{it} = c_i + \mathbf{x}'_{it}\beta + \theta S_{it} + \varepsilon_{it},$$

where $t = 1, 2$, and $S_{it} = 0$ is period 1 and $S_{it} = 1$ in period 2.

The treatment effect would be $E[\Delta y_i | \Delta \mathbf{x}_i = 0]$. So, taking the first difference, we find

$$\Delta y_i = \theta + (\Delta \mathbf{x}_{it})'\beta + u_i$$

For $\Delta \mathbf{x}_i = 0$, the treatment effect is θ.

The Within- and Between-Groups Estimator

Consider now three models we can consistently estimate by OLS:

$$\begin{aligned} \text{Pooled regression model: } y_{it} =& \alpha + \mathbf{x}_{it}\beta + \varepsilon_{it} \\ \text{Group means model: } \bar{y}_i =& \alpha + \bar{\mathbf{x}}_i\beta + \bar{\varepsilon}_i \\ \text{Deviation from group means: } y_{it} - \bar{y}_i =& (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\beta + \varepsilon_{it} - \bar{\varepsilon}_i \end{aligned}$$

For convenience, we write the last model as

$$\ddot{y} = \mathbf{x}'_{it}\beta + \ddot{\varepsilon}_{it}$$

The Within- and Between-Groups Estimator

Consider then the matrices of sums of squares and cross products that would be used in each case:

$$S_{xx}^{total} = \sum_{i=1}^{n}\sum_{t=1}^{T}(\mathbf{x}_{it} - \bar{\bar{\mathbf{x}}})(\mathbf{x}_{it} - \bar{\bar{\mathbf{x}}})' \text{ and } S_{xy}^{total} = \sum_{i=1}^{n}\sum_{t=1}^{T}(\mathbf{x}_{it} - \bar{\bar{\mathbf{x}}})(\mathbf{y}_{it} - \bar{\bar{\mathbf{y}}})'$$

where $\bar{\bar{\mathbf{x}}}$ and $\bar{\bar{\mathbf{y}}}$ are overall means.

The Within- and Between-Groups Estimator

$$S_{xx}^{between} = \sum_{i=1}^{n} T(\mathbf{x}_i - \bar{\bar{\mathbf{x}}})(\mathbf{x}_i - \bar{\bar{\mathbf{x}}})' \text{ and } S_{xy}^{between} = \sum_{i=1}^{n} T(\mathbf{x}_i - \bar{\bar{\mathbf{x}}})(\mathbf{y}_i - \bar{\bar{\mathbf{y}}})'$$

and

$$S_{xx}^{within} = \sum_{i=1}^{n}\sum_{t=1}^{T} (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)' \text{ and } S_{xy}^{within} = \sum_{i=1}^{n}\sum_{t=1}^{T} (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)(\mathbf{y}_{it} - \bar{\mathbf{y}}_i)'$$

Notice that for this last sums, because the data are in deviations already, the means of $(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)$ and $(\mathbf{y}_{it} - \bar{\mathbf{y}}_i)$ are zero.

The Within- and Between-Groups Estimator

Moreover, it can be shown that

$$S_{xx}^{total} = S_{xx}^{between} + S_{xx}^{within} \text{ and } S_{xy}^{total} = S_{xy}^{between} + S_{xy}^{within}$$

Therefore, there are three possible LS estimators of β:

$$\mathbf{b}^{total} = [S_{xx}^{total}]^{-1} S_{xy}^{total} = [S_{xx}^{between} + S_{xx}^{within}]^{-1} [S_{xy}^{between} + S_{xy}^{within}]$$

$$\mathbf{b}^{between} = [S_{xx}^{between}]^{-1} S_{xy}^{between}$$

$$\mathbf{b}^{within} = [S_{xx}^{within}]^{-1} S_{xy}^{within}$$

The Within- and Between-Groups Estimator

From the preceding expressions, we have

$$S_{xy}^{between} = S_{xx}^{between}\mathbf{b}^{between} \text{ and } S_{xy}^{within} = S_{xx}^{within}\mathbf{b}^{within}$$

Inserting these expressions for $\mathbf{b}^{total}$,

$$\mathbf{b}^{total} = \mathbf{F}^{within}\mathbf{b}^{within} + \mathbf{F}^{between}\mathbf{b}^{between}$$

where $\mathbf{F}^{within} = [S_{xx}^{between} + S_{xx}^{within}]^{-1} S_{xx}^{within} = \mathbf{I} - \mathbf{F}^{between}$.

We see that the LS estimator is a **matrix weighted average** of the within- and the between-groups estimators.

The Fixed Effects Model

Consider again the general model:

$$y_{it} = \alpha + \mathbf{x}_{it}'\beta + \varepsilon_{it}, i = 1, \ldots, n; t = 1, \ldots, T_i$$

$$\begin{aligned} E[\varepsilon_{it}|\mathbf{x}_{i1}, \mathbf{x}_{i2}, \ldots, \mathbf{x}_{iT_i}] =& 0 \\ E[\varepsilon_{it}\varepsilon_{js}|\mathbf{x}_{i1}, \mathbf{x}_{i2}, \ldots, \mathbf{x}_{iT_i}] =& \sigma_\varepsilon^2 \text{ if } i = j \text{ and } t = s \text{ and } = 0 \text{ if } i \neq j \text{ or } t \neq s \end{aligned}$$

In the fixed effects model, the omitted effects, c_i, can be arbitrarily correlated with the included variables.

In a generic form,

$$E[c_i|\mathbf{x}_{i1}, \mathbf{x}_{i2}, \ldots,] = E[c_i|\mathbf{X}_i] = h(\mathbf{X}_i)$$

This is what signifies the fixed effects model, not that any variable is fixed in this context and random elsewhere, despite the terminology.

The Fixed Effects Model

Let's also assume that $Var[c_i|\mathbf{X}_i]$ is constant and all observations c_i and c_j are independent.

The formulation implies that the heterogeneity across groups is captured in the constant term:

$$y_{it} = \alpha_i + \mathbf{x}'_{it}\beta + \varepsilon_{it}$$

Each α_i can be treated as an unknown parameter to be estimated.

Least Squares Estimation

Let $\mathbf{y}_i$ and $\mathbf{X}_i$ denote the T observations for the ith unit, let $\mathbf{i}$ be a $T \times 1$ column vector of ones, and let ε_i be the associated $T \times 1$ vector of disturbances.

Then,

$$\mathbf{y}_i = \mathbf{i}\alpha_i + \mathbf{x}'_{it}\beta + \varepsilon_i$$

We can represent the model above as

$$\mathbf{y}_i = \mathbf{D}\alpha + \mathbf{X}\beta + \varepsilon$$

where $D = [\mathbf{d}_1, \mathbf{d}_2], \ldots, \mathbf{d}_n]$, and d_i is a dummy vector indicating the ith unit.

This model is referred to as the **least squares dummy variable (LSDV) model**.

The LSDV Model

We can estimate this model by least squares, with $n + K$ parameters to be estimated.

If n is thousands, as it often is the case, then treating it as an ordinary regression will be extremely cumbersome.

Fortunately, we can resort to the FWL theorem, reducing the size of the computation by a large amount.

The LSDV Model

We can write the LS estimator of β as

$$\mathbf{b}_{LSDV} = \left[\mathbf{X}'\mathbf{M_D}\mathbf{X}\right]^{-1}\left[\mathbf{X}'\mathbf{M_D}\mathbf{y}\right] = \left[(\mathbf{X}'\mathbf{M_D})(\mathbf{M_D}\mathbf{X})\right]^{-1}\left[(\mathbf{X}'\mathbf{M_D})(\mathbf{M_D}\mathbf{y})\right]$$

where $\mathbf{M_D} = \mathbf{I}_{nT} - \mathbf{D}(\mathbf{D}'\mathbf{D})^{-1}\mathbf{D}'$.

This amounts to a LS regression using the transformed data $\mathbf{M_D}\mathbf{X}$ and $\mathbf{M_D}\mathbf{y}$.

It can be shown that $\mathbf{b}_{LSDV} = \mathbf{b}^{within}$.

The LSDV Model

In terms of the transformed data, then,

$$\mathbf{b}_{LSDV} = \left[\sum_{i=1}^{n} (\mathbf{M^0X_i})'(\mathbf{M^0X_i})\right]^{-1} \left[\sum_{i=1}^{n} (\mathbf{M^0X_i})'(\mathbf{M^0y_i})\right]$$
$$= \left[\sum_{i=1}^{n} \ddot{\mathbf{X}}_i'\ddot{\mathbf{X}}_i\right]^{-1} \left[\sum_{i=1}^{n} \ddot{\mathbf{X}}_i'\ddot{\mathbf{y}}_i\right] = (\ddot{\mathbf{X}}'\ddot{\mathbf{X}})^{-1}\ddot{\mathbf{X}}'\ddot{\mathbf{y}}$$

We can recover the dummy variable coefficients from the other normal equation in the partition regression,

$$\mathbf{D'Da} + \mathbf{D'Xb}_{LSDV} \text{ or } \mathbf{a} = [\mathbf{D'D}]^{-1}\mathbf{D'}(\mathbf{y} - \mathbf{D'Xb}_{LSDV})$$

This implies that for each i,

$$a_i = \bar{\mathbf{y}}_i - \bar{\mathbf{x}}_i'\mathbf{b}_{LSDV}$$

The LSDV Model

The appropriate estimator for the asymptotic covariance matrix of **b** is

$$Est.Asy.Var[\mathbf{b}_{LSDV}] = s^2 \left[\sum_{i=1}^{n} \ddot{\mathbf{X}}_i' \ddot{\mathbf{X}}_i \right]^{-1}$$

where

$$s^2 = \frac{(\ddot{\mathbf{y}} - \ddot{\mathbf{X}}\mathbf{b}_{LSDV})'(\ddot{\mathbf{y}} - \ddot{\mathbf{X}}\mathbf{b}_{LSDV})}{nT - n - K}$$

The LSDV Model

The ith residual in this computation is

$$e_i = y_{it} - \mathbf{x}_{it}'\mathbf{b}_{LSDV} - a_i = y_{it} - \mathbf{x}_{it}'\mathbf{b}_{LSDV} - (\bar{\mathbf{y}}_i - \bar{\mathbf{x}}_i'\mathbf{b}_{LSDV})$$
$$= (y_{it} - \bar{\mathbf{y}}_i) - (\mathbf{x}_{it} - \bar{\mathbf{x}})'\mathbf{b}_{LSDV}$$

Thus, the numerator in s^2 is exactly the sum of squared residuals using the LS slopes and the data in group mean deviation form.

Also, we need to adjust the denominator to $nT - K$ instead of $nT - n - K$.

The LSDV Model

For the individual effects, $a_i = \bar{y}_i - \bar{\mathbf{x}}_i' \mathbf{b}_{LSDV}$,

$$Asy.Var[a_i] = \frac{\sigma_\varepsilon^2}{T} + \bar{\mathbf{x}}_i' \{Asy.Var[\mathbf{b}]\} \bar{\mathbf{x}}_i$$

so a simple estimator based on s^2 can be computed.

Notice that $Asy.Var[a_i]$ does not converge to zero as n grows. Thus, the constant term estimators in the fixed effects model are **not** consistent estimators of α_i.

- Each α_i is estimated using only T observations.

Moreover, notice that **all** time-invariant variables will be absorbed by the model (a major drawback).

Testing for the Significance of the Group Effect

If we are interested in testing for the presence of heterogeneity across groups, the, we can test the hypothesis that the constant terms are all equal with an F test.

Under the null hypothesis of equality, the efficient estimator is the pooled least squares.

The F ratio used for this test is

$$F(n-1, nT-n-K) = \frac{(R^2_{LSDV} - R^2_{pooled})/n-1}{(1-R^2_{LSDV})/nT-n-K}$$

Fixed Time and Group Effects

The LSDV approach can be extended to include a time-specific effect as well.

One way to formulate the extended model is simply to add the time effect, as in

$$y_{it} = \mathbf{x}'_{it}\beta + \alpha_i + \delta_t + \varepsilon_{it}$$

This model simply includes an additional $T-1$ dummy variables to the model.

We can describe a generalized differences-in-differences approach based on the model above (the infamous Two-Way Fixed Effects)

Fixed Time and Group Effects

An alternative extension would be the inclusion of a time trend and/or unit-specific linear time trends (linear trends that vary by unit):

$$y_{it} = \mathbf{x}'_{it}\beta + t.\alpha_i + t + \varepsilon_{it}$$

Random Effects Model

The fixed effects model allows the unobserved individual effects to be correlated with the included variables.

If the individual effects are uncorrelated with the regressors, then it might be appropriate the model the individual-specific constant terms as randomly distributed across cross-sectional units.

Consider the reformulation of the model,

$$y_{it} = \mathbf{x}'_{it}\beta + (\alpha + u_i) + \varepsilon_{it},$$

where there are K regressors (including a constant) and now the single constant term is the mean of the unobserved heterogeneity, $E[\mathbf{z}'_i\alpha]$.

- Recall that $u_i = [\mathbf{z}'_i\alpha - E(\mathbf{z}'_i\alpha)]$.

Random Effects Model

We continue to assume strict exogeneity:

$$
\begin{aligned}
E[\varepsilon_{it}|\mathbf{X}_i] =& E[u_i|\mathbf{X}_i] = 0 \\
E[\varepsilon_{it}^2|\mathbf{X}_i] =& \sigma_\varepsilon^2 \\
E[u_i^2|\mathbf{X}_i] =& \sigma_u^2 \\
E[\varepsilon_{it}u_j|\mathbf{X}_i] =& 0 \text{ for all } i, t, \text{ and } j \\
E[\varepsilon_{it}\varepsilon_{js}|\mathbf{X}_i] =& 0 \text{ if } i \neq j \text{ or } t \neq s \\
E[u_i u_j|\mathbf{X}_i] =& 0 \text{ if } i \neq j
\end{aligned}
$$

Random Effects Model

As before, let's also represent the model in blocks of T observations for group i, $\mathbf{y}_i$, $\mathbf{X}_i$, u_i, and ε_i.

For these T observations, let

$$\eta_{it} = \varepsilon_{it} + u_i$$

and

$$\eta_i = [\eta_{i1}, \eta_{i2}, \ldots, \eta_{iT}]'$$

In view of this form of η_{it}, we have what is often calld an **error components model**.

Random Effects Model

For this model,

$$
\begin{aligned}
E[\eta_{it}|\mathbf{X}_i] =& \sigma_\varepsilon^2 + \sigma_u^2 \\
E[\eta_{it}\eta_{js}|\mathbf{X}_i] =& \sigma_u^2, t \neq s \\
E[\eta_{it}\eta_{js}|\mathbf{X}_i] =& 0, \text{ for all } t \text{ and } s, \text{ if } i \neq j
\end{aligned}
$$

For the T observations for unit i, let $\mathbf{\Sigma} = E[\eta_i \eta_i' | \mathbf{X}_i]$

Random Effects Model

Then,

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_\varepsilon^2 + \sigma_u^2 & \sigma_u^2 & \sigma_u^2 & \dots & \sigma_u^2 \\ \sigma_u^2 & \sigma_\varepsilon^2 + \sigma_u^2 & \sigma_u^2 & \dots & \sigma_u^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_u^2 & \sigma_u^2 & \sigma_u^2 & \dots & \sigma_\varepsilon^2 + \sigma_u^2 \end{bmatrix} = \sigma_\varepsilon^2 \mathbf{I}_T + \sigma_u^2 \mathbf{i}_T \mathbf{i}_T'$$

Random Effects Model

Because observations i and j are independent, the disturbance covariance matrix for the full nT observations is

$$\boldsymbol{\Omega} = \begin{bmatrix} \boldsymbol{\Sigma} & \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma} & \mathbf{0} & \dots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \dots & \boldsymbol{\Sigma} \end{bmatrix} = \mathbf{I}_n \otimes \boldsymbol{\Sigma}$$

The Least Squares Estimation

Consider again random effects the model:

$$y_{it} = \alpha + \mathbf{x}'_{it}\beta + u_i + \varepsilon_{it},$$

with the strict exogeneity assumptions and the covariance matrix as detailed in the preceding discussion.

Given that disturbances are correlated, because observations are correlated across time within a group (though not across groups), the generalized regression model above fits into the framework we developed in the previous chapter, the GLS model.

The Least Squares Estimation

Notice that we have also discussed four other methods that consistently estimates β:

1. OLS estimator
2. the LSDV estimator
3. First difference estimator
4. Group means (betweem group) estimator

However, they are all inefficient! (why?)

Generalized Least Squares

The GLS estimator of the slope parameters is

$$\hat{\beta} = (\mathbf{X'\Omega^{-1}X})^{-1}\mathbf{X'\Omega^{-1}y} = \left[\sum_{i=1}^{n} \mathbf{X_i'\Sigma^{-1}X_i}\right]^{-1} \left[\sum_{i=1}^{n} \mathbf{X_i'\Sigma^{-1}y_i}\right]$$

As before, to compute $\hat{\beta}$, w transform the data and use OLS with the transformed data.

This requires $\Omega^{-1/2} = [\mathbf{I}_n \otimes \mathbf{\Sigma}]^{-1/2} = \mathbf{I}_n \otimes \mathbf{\Sigma}^{-1/2}$

We have that

$$\mathbf{\Sigma}^{-1/2} = \left[\mathbf{I}_T - \frac{\theta}{T}\mathbf{i}_T\mathbf{i}_T'\right],$$

where $\theta = 1 - \frac{\sigma_\varepsilon^2}{\sqrt{\sigma_\varepsilon^2 + T\sigma_u^2}}$.

Generalized Least Squares

The transformation of $\mathbf{y}_i$ and $\mathbf{X}_i$ for GLS is therefore,

$$\mathbf{\Sigma}^{-1/2}\mathbf{y}_i = \frac{1}{\sigma_\varepsilon^2}\begin{bmatrix} y_{i1} - \theta\bar{\mathbf{y}}_i \\ y_{i2} - \theta\bar{\mathbf{y}}_i \\ \vdots \\ y_{iT} - \theta\bar{\mathbf{y}}_i \end{bmatrix}$$

and likewise for the rows of $\mathbf{X}_i$.

Notice that the LSDV model is the same as above, but using $\theta = 1$.

Feasible Generalized Least Squares Estimation of the Random Effects Model when $\mathbf{\Sigma}$ is Unknown

If the variance components are known, GLS can be computed as shown earlier.

Or course, this is unlikely, so as usual, we must first estimate the disturbance variances and then use a FGLS procedure.

A heuristic approach to estimation of the variance components is as follows:

$$y_{it} = \alpha + \mathbf{x}'_{it}\beta + u_i + \varepsilon_{it},$$

and

$$\bar{y}_i = \alpha + \bar{\mathbf{x}}'_i\beta + u_i + \bar{\varepsilon}_i$$

Therefore, taking deviations from group means removes the heterogeneity.

Feasible Generalized Least Squares Estimation of the Random Effects Model when $\mathbf{\Sigma}$ is Unknown

We can, thus, estimate β by the LSDV estimator, which is consistent and unbiased in most cases, and use its residuals,

$$s_e^2(i) = \frac{\sum_{t=1}^{T} (e_{it} - \bar{e}_i)^2}{T - K - 1}$$

Notice that $\bar{e}_i = 0$ in the LSDV model, but we carry it out to stress that we are actually estimating $E[\varepsilon_{it}]$.

We have n such estimators, so we average them to obtain

$$\bar{s}_e^2 = \frac{1}{n}\sum_{i=1}^{n} s_e^2(i) = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{\sum_{t=1}^{T}(e_{it} - \bar{e}_i)^2}{T - K - 1}\right] = \frac{\sum_{i=1}^{n}\sum_{t=1}^{T}(e_{it} - \bar{e}_i)^2}{nT - nK - n}$$

Feasible Generalized Least Squares Estimation of the Random Effects Model when $\boldsymbol{\Sigma}$ is Unknown

The degrees of freedom correction in $\bar{s}_e^2$ is excessive, because it assumes that α and β are reestimated for each i.

To account for that, we have

$$\hat{\sigma}_\varepsilon^2 = \frac{\sum_{i=1}^n \sum_{t=1}^T (e_{it} - \bar{e}_i)^2}{nT - n - K} = s_{LSDV}^2$$

Feasible Generalized Least Squares Estimation of the Random Effects Model when $\boldsymbol{\Sigma}$ is Unknown

It remanis to estimate σ_u^2.

Returning to the original model,

$$y_{it} = \alpha + \mathbf{x}_{it}'\beta + u_i + \varepsilon_{it},$$

If we estimate the above model using the pooled regression model, with only a single overall constant, we would find

$$\text{plim } s^2_{pooled} = \text{plim } \frac{\mathbf{e}'\mathbf{e}}{nT - K - 1} = \sigma_\varepsilon^2 + \sigma_u^2$$

because $\mathbf{b}_{pooled}$ is a consistent estimator of β.

Therefore, the estimator for σ_u^2 would be

$$\hat{\sigma}_u^2 = s^2_{pooled} - s^2_{LSDV}$$

Robust Inference and Feasible Generalized Least Squares

The GLS estimator is

$$\hat{\beta} = (\mathbf{X'\Omega^{-1}X})^{-1}\mathbf{X'\Omega^{-1}y} = \left[\sum_{i=1}^{n} \mathbf{X_i'\Sigma^{-1}X_i}\right]^{-1}\left[\sum_{i=1}^{n} \mathbf{X_i'\Sigma^{-1}y_i}\right]$$

and the feasible GLS estimator is

$$\hat{\hat{\beta}} = \beta + \left[\sum_{i=1}^{n} \mathbf{X_i'\hat{\Sigma}^{-1}X_i}\right]^{-1}\left[\sum_{i=1}^{n} \mathbf{X_i'\hat{\Sigma}^{-1}}\varepsilon_i\right]$$

Following the approach used in earlier applications, the estimated robust covariance matrix would be

$$Est.Asy.Var[b] = \left[\sum_{i=1}^{n} \mathbf{X_i'\hat{\Sigma}^{-1}X_i}\right]^{-1}\left[\sum_{i=1}^{n} \left(\mathbf{X_i'\hat{\Sigma}^{-1}e}_i\right)\left(\mathbf{X_i'\hat{\Sigma}^{-1}e}_i\right)'\right]\left[\sum_{i=1}^{n} \mathbf{X_i'\hat{\Sigma}^{-1}X_i}\right]$$

Hausman's Specification Test for the Random Effects Model

At various points, we have made the distinction between fixed and random effects models.

- Which one should we use?

The dummy variable approach is costly in terms of degrees of freedom lost.

On the other hand, in the likely case of the individual effects being correlated with the other regressors, the random effects approach is **inconsistent**.

The specification test devised by Hausman (1978) is used to test for the orthogonality of the common effects and the regressors.

Hausman's Specification Test for the Random Effects Model

The test is based on the idea that under the hypothesis of no correlation, both LSDV and FGLS estimators are consistent, but the LSDV is inefficient, whereas under the alternative, LSDV is consistent but FGLS is not.

Under H_0, both $\mathbf{b}_{LSDV}$ and $\hat{\beta}_{pooled}$ are consistent, so the two estimates should not differ systematically.

The chi-squared test based on the Wald criterion is

$$\mathbf{W} = \chi^2[K-1] = [\mathbf{b}_{LSDV} - \hat{\beta}_{pooled}]'[Var(\mathbf{b}_{LSDV} - \hat{\beta}_{pooled})]^{-1}[\mathbf{b}_{LSDV} - \hat{\beta}_{pooled}]$$

Under the null hypothesis, $\mathbf{W}$ has a limiting chi-squared distribution with $K-1$ degrees of freedom.

Hausman's Specification Test for the Random Effects Model

We need to find $Var(\mathbf{b}_{LSDV} - \hat{\beta}_{pooled})$. Hausman's essential result is that *the covariance of an efficient estimator with its difference from an inefficient estimator is zero.*

Thus, we can show that $Var(\mathbf{b}_{LSDV} - \hat{\beta}_{pooled}) = Var(\mathbf{b}_{LSDV}) - Var(\hat{\beta}_{pooled})$

Hausman's Specification Test for the Random Effects Model

One drawback is that we cannot guarantee that the difference of the two covariance matrices will be positive definite in a finite sample.

An asymptotically equivalent test statistic is

$$\mathbf{H} = (\mathbf{b}_{LSDV} - \mathbf{b}_{means})'[Asy.Var[\mathbf{b}_{LSDV}] + Asy.Var[\mathbf{b}_{means}]]^{-1}(\mathbf{b}_{LSDV} - \mathbf{b}_{means})$$

Applying the Fixed Effects Model

We can use fixed effects for other data structures to restrict comparisons to **within unit variation**.

For example:

1. Matched pairs (such as Twin fixed effects to control for unobserved effects of family background)
2. Cluster fixed effects in hierarchical data (such as school fixed effects to control for unobserved effects of school)

Effects of Birth Endowments and Human Capital Accumulation - Twin Fixed Effects Approach

Is being born with better health good for subsequent human capital?

- Facilitates learning
- Better health increases education productivity
- Better education can affect labor market and welfare outcomes

Black *et al* (2007) use twin individuals and estimate

$$y_{it} = \alpha + \beta health_{it} + \gamma_j + \varepsilon_{it}$$

where γ_j is the twin-fixed effects (FE). It absorbs all omitted factors that are common to twins born from the same mother.

Identifying assumption: conditional on twin FE, the only relevant difference between twins is the birth health.

Results in Black et al (2007)

Regression Results: Twins Sample Coefficient on LN (Birth Weight)

Dependent variable	Singleton sample: OLS	Singleton sample: Family fixed effects	Twins sample: OLS	Twins sample: Twin fixed effects
One-year mortality	−123.46** (1.71)	−186.71** (.69)	−279.64** (9.12)	−41.10** (7.64)
N	1,253,546		33,366	
Five minute APGAR score	.73** (.01)	1.08** (.01)	1.46** (.06)	.35** (.07)
N	674,577		21,580	
Height (males only)	11.03** (.11)	7.33** (.12)	7.48** (.55)	5.68** (.56)
N	203,741		5,382	
BMI (males only)	−6.19 (7.67)	−22.22 (15.23)	.56** (.23)	1.12** (.30)
N	203,378		5,372	
Underweight	−.09** (.004)	−.07** (.01)	−.07** (.02)	−.11** (.04)
N	203,378		5,372	
Overweight	.08** (.01)	.08** (.01)	.03 (.02)	.09** (.04)
N	203,378		5,372	
IQ (males only)	.91** (.03)	.58** (.04)	.48** (.14)	.62** (.18)
N	184,045		4,920	
High school completion	.16** (.01)	.04** (.01)	.07** (.02)	.09** (.04)
N	536,020		13,106	
Full-time work	.17** (.004)	.21** (.01)	.29** (.02)	.03 (.05)
N	368,582		10,388	
ln(earnings) FT	.09** (.01)	.08** (.01)	.09** (.03)	.12** (.06)
N	239,906		5,952	
ln(birth weight of first child)	.25** (.01)	.13** (.01)	.18** (.04)	.15** (.06)
N	63,842		1,862	

Problems that Fixed Effects Do Not Solve

Consider the following regression model:

$$y_{it} = \mathbf{x}_{it}\beta + c_i + \varepsilon_{it}$$

where y_{it} is murder rate and $\mathbf{x}_{it}$ is police spending per capita
What happens when we regress y on x and city fixed effects?

- $\hat{\beta}_{FE}$ is inconsistent unless strict exogeneity conditional on c_i holds
- Implies ε_{it} is uncorrelated with past, current and future regressors

Most common violations:

1. Time-varying omitted variables
2. Simultaneity

Fixed effects do not obviate need for good research design!

Fixed Effects and Differences-in-Differences

"Diff-in-Diff" is a common panel technique used for identification.

In the basic setup, we have:

- Two periods, *pre* and *post*, where post period identified by the dummy variable, $Post_t$
- Two groups, Treatment and Control, where the treatment group identified by the dummy variable, $Treat_i$.

Fixed Effects and Differences-in-Differences

Diff-in-Diff Specification:

$$y_{it} = \beta_0 + \beta_1 Post_t + \beta_2 Treat_i + \delta Post_t \times Treat_i + \varepsilon_{it}$$

- δ is the differences-in-differences”estimator.
- It is also called an “average treatment effect”

Where does the term "diff-in-diff" come from?

"Diff-in-Diff" takes two differences:

1. Pre vs Post
2. Treatment vs Control

$$y_{it} = \beta_0 + \beta_1 Post_t + \beta_2 Treat_i + \delta Post_t \times Treat_i + \varepsilon_{it}$$

Write out predictions, their difference, then the difference in differences:

	Before	After	After - Before
Control	β_0	$\beta_0 + \beta_1$	β_1
Treatment	$\beta_0 + \beta_2$	$\beta_0 + \beta_1 + \beta_2 + \delta$	$\beta_1 + \delta$
Control - Treatment	β_2	$\beta_2 + \delta$	δ

Where does the term "diff-in-diff" come from?

We have the "diff-in-diff" specification below:

$$y_{it} = \beta_0 + \beta_1 Post_t + \beta_2 Treat_i + \delta Post_t \times Treat_i + \varepsilon_{it}$$

Treatment group might naturally differ from control: β_2

Post might naturally differ from Pre: β_1

Getting rid of both yields δ, the average treatment effect

Diff-Diff Identifying Assumption

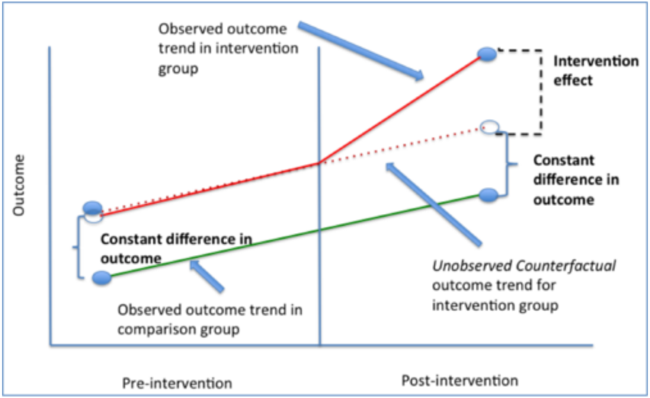
The assumption for identification of the average treatment effect from δ is: *in the absence of a real treatment, control and treated units would have followed the same trends.*

This assumption is called “**the common trends assumption**” or “**the parallel trends assumption**

It is untestable by definition.

An important implication is that treated and untreated units need no to be the same before policy adoption.

Diff-Diff Identifying Assumption



Threats to DID Assumptions

There are several reasons why the parallel assumption could fail:

- Lack of parallel trends
- Differential attrition in treated and untreated units, which may lead to compositional changes.

It is impossible to test the common trends assumptions, but:

- We can check pre-trends
- If trends are similar before policy, then they would likely have continued to be similar in the post-period absent a real treatment.
- Hence, it is important to have several periods before policy to really convince about the plausibility of the identifying assumption.

A More General DID Framework

In practice, we have more than two periods, although the treatment occurs after a given date.

The identifying assumption could be valid conditional on certain variables

It could be interesting to include both time and unit fixed effects.

So, a more general Diff-in-Diff specification is:

$$y_{it} = \beta_0 + \delta Post_t \times Treat_i + \beta_1 \mathbf{X}_{it} + \lambda_i + t_t + \varepsilon_{it}$$

where λ_i are unit fixed effects and t_t are time fixed effects.

Event-Study Specifications and Assumption in DID

How to check for pre-trends to convincingly evaluate the parallel trends assumption?

Estimate the differences-in-differences for each period before and after policy adoption:

$$y_{it} = \beta_0 + \sum \delta_k^{pre} D_k^{pre} \times Treat_i + \sum \delta_k^{post} D_k^{post} \times Treat_i + \lambda_i + t_t + \varepsilon_{it}$$

If the identifying assumption is valid, then $\delta_k^{pre} = 0$ for all k before policy adoption

That is, the trends before policy are the same.

Other Important Extensions in DID

Differences-in-Differences do not nee to be *only* differences between units over time

Interesting applications use differences within and between different groups.

- Examples: Differences between black and white in different states or countries

There may be more comparison groups and implement differences-in-differences-in-differences:

- Comparison between black and white in different states or countries before and after policy adoption

Duflo (2001). *Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence from an Unusual Policy Experiment.* **American Economic Review**

Question and Motivation

More school infraestructure, more schooling?

Educational level in turns affect wages?

What is the economic return to education?

Program Description

1973-1978: The Indonesian government built 61,000 schools equivalent to one school per 500 children between 5 and 14 years old

The enrollment rate increased from 69% to 85% between 1973 and 1978

The number of schools built in each region depended on the number of children out of school in those regions in 1972, before the start of the program.

Treatment Effect Identification

There are 2 sources of variations in the intensity of the program for a given individual:

1. **By region**: There is variation in the number of schools received in each region.
2. **By age**: Children who were older than 12 years in 1972 did not benefit from the program, whereas the younger a child was 1972, the more it benefited from the program — because she spent more time in the new schools.

Sources of Data

1995 population census. Individual-level data on:

1. Birth date
2. 1995 salary level
3. 1995 educational level

The intensity of the building program in the birth region of each person in the sample.

Sample: men born between 1950 and 1972.

A First Estimation of the Impact

Step 1: Simplify the intensity of the program and the groups of children affected by the program:

- High or low areas with school construction
- Young cohort of children who benefited/Older cohort of children who did not benefit

Look at the Average Years of Schooling

Intensity of the Building Program

Age in 1974	High	Low	
2-6 (young cohort)	8.49	9.76	
12-17 (older cohort)	8.02	9.4	
Difference	**0.47**	**0.36**	**0.12 DD** **(0.089)**

Look at the Average Years of Schooling

Intensity of the Building program

Age in 1974	High	Low	**Difference**
2-6 (young cohort)	8.49	9.76	**-1.27**
12-17 (older cohort)	8.02	9.4	**-1.39**
			0.12 DD **(0.089)**

Placebo or Falsification Test

Idea: Look for 2 groups whom you know did not benefit, compute a DID, and check whether the estimated effect is 0.

Intensity of the Building Program

Age in 1974	High	Low	
12-17	8.02	9.40	
18-24	7.70	9.12	
Difference	**0.32**	**0.28**	**0.034 DD** **(0.098)**

Formal Regression Results

Step 2:

$$S_{ijk} = c + \alpha_j + \beta_k + \gamma P_j \times T_i + C_j \times T_i + \varepsilon_{ijk}$$

where:

- S_{ijk} educational level of the person i in region j in cohort k
- $P_j = 1$ if the person was born in a region with a high program intensity
- $T_i = 1$ if the person belongs to the "young" cohort
- C_j dummy variable for region j
- β_k cohort fixed effects
- α_j district of birth

The key **differences-in-differences parameter** is γ

Formal Regression Results

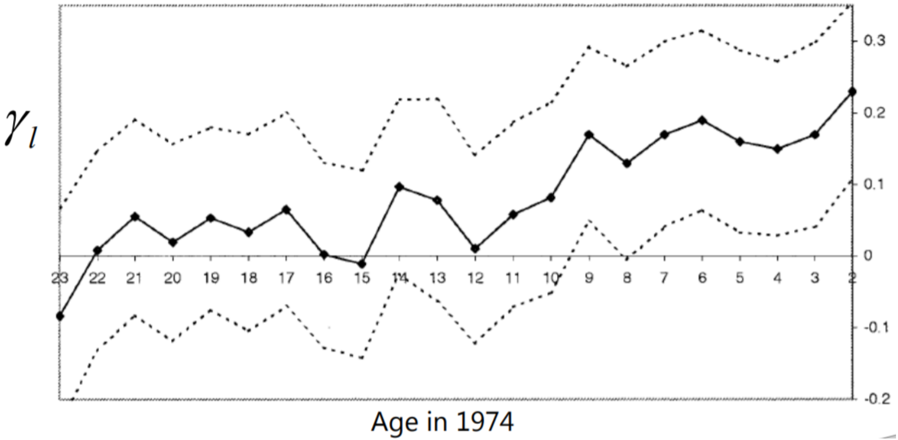
Step 3: We will use the intensity of the program in each region.

We estimate the effect of the program for each cohort separately:

$$S_{ijk} = c + \alpha_j + \beta_k + \sum_{i=2}^{23} \gamma P_j \times d_i + C_j \times T_i + \varepsilon_{ijk}$$

where d_i a dummy variable for belonging to cohort i

Program Effect per Cohort



Effects on Salary

	Log(wages)		
	Level of program in region of birth		
	High (4)	Low (5)	Difference (6)
Panel A: Experiment of Interest			
Aged 2 to 6 in 1974	6.61	6.73	−0.12
	(0.0078)	(0.0064)	(0.010)
Aged 12 to 17 in 1974	6.87	7.02	−0.15
	(0.0085)	(0.0069)	(0.011)
Difference	−0.26	−0.29	0.026
	(0.011)	(0.0096)	(0.015)
Panel B: Control Experiment			
Aged 12 to 17 in 1974	6.87	7.02	−0.15
	(0.0085)	(0.0069)	(0.011)
Aged 18 to 24 in 1974	6.92	7.08	−0.16
	(0.0097)	(0.0076)	(0.012)
Difference	0.056	0.063	0.0070
	(0.013)	(0.010)	(0.016)

Conclusions

Results: For each school built per 1000 students;

- The average educational achievement increase by 0.12-0.19 years
- The average salaries increased by 2.6 – 5.4%

Making sure the DD estimation is accurate:

- A placebo DD gave 0 estimated effect
- Use various alternative specifications
- Check that the impact estimates for each age cohort make sense

Cheng and Hoekstra (2013). *Does Strengthening Self-Defense Law Deter Crime or Escalate Violence?: Evidence from Expansions to Castle Doctrine.* **Journal of Human Resources**

Summary

Cheng and Hoekstra (2013) are interested in whether expansions to “castle doctrine statutes” at the state level increase or decrease gun violence

Prior to these expansions, English common law principle required “duty to retreat” before using lethal force against an assailant except when the assailant is an intruder in the home

- The home is one’s “castle” - hence, “castle doctrine”
- When intruders threatened the victim in the home, the duty to retreat was waived and lethal force in self-defense was allowed

Castle Doctrine Law Explained

In 2005, Florida passed a law that expanded self-defense protections beyond the house.

From 2000 to 2010, 21 states explicitly put castle doctrine into statute, and (more importantly) extended it to places outside the home.

- In other words, 21 states removed the duty to retreat in specified circumstances

Other changes:

- Presumption of reasonable fear is added
- Civil liability for those acting under the law is removed

Economic Theory

Workers supply legal or illegal labor

- Costs: foregone wages, risk of arrest, risk of conviction, penalties conditional on conviction
- Benefits: valuable goods

Castle doctrine expansions lowered the (expected) cost of killing someone in self-defense

If people are rational, then lowering the price of lethal self-defense should increase lethal homicides

Cheng and Hoekstra's Identification Strategy

Research design:

- Estimate the state-level causal effect: what would've happened to these same states had they not passed the law?
- Compare the changes in outcomes after castle doctrine law adoption to changes in the outcomes in other states in the same region of the country

Estimation: Panel fixed effects estimation

$$y_{it} = \beta_0 + \delta CDL_{it} + \beta_1 \mathbf{X}_{it} + \lambda_i + t_t + \varepsilon_{it}$$

CDL is a dummy variable equaling 1 if the state has a castle doctrine law and zero otherwise.

Data

FBI Uniform Crime Reports (2000-2010):

- State-level crime rates, or “offenses per 100,000 population”

Homicide categories:

- Total homicides - murder plus non-negligent manslaughter
- Justifiable homicides by private citizens

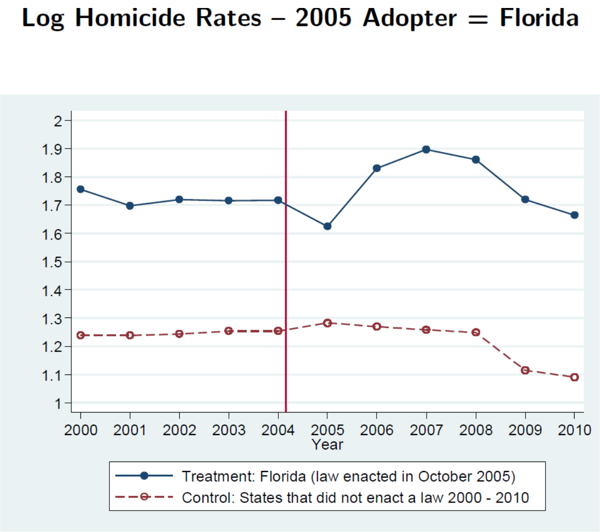
Region-by-Year Fixed Effects

Most of their specifications will include region dummies interacted with year dummies or “region-by-year fixed effects”

Absent passing castle doctrine laws, outcomes in these 21 states would have changed similar to other states in their same region.

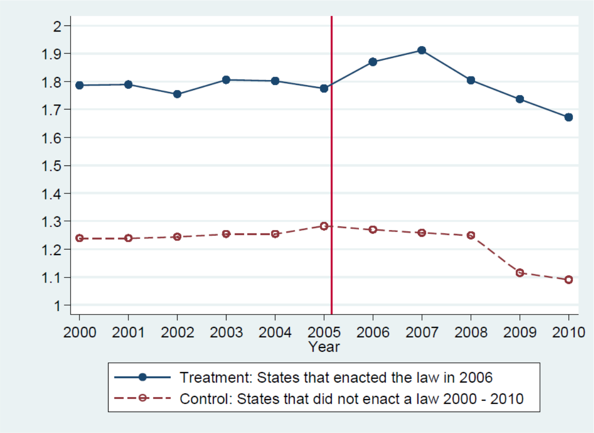
- By including “region-by-year fixed effects”, they are arguing that unobserved changes in crime are running “parallel”to the treatment states within region over time
- Need not hold across regions since the across region variation is not being used in this analysis due to the saturation of the model with “region-by-year fixed effects”

Testing the identification strategy



Testing the identification strategy

Log Homicide Rates – 2006 Adopter (13 states)



Results

Homicide – OLS

	1	2	3	4	5	6
Panel A: Log Homicide Rate (OLS - Weighted)						
Castle Doctrine Law	0.0801**	0.0946***	0.0937***	0.0875**	0.0985***	0.100**
	(0.0342)	(0.0279)	(0.0290)	(0.0337)	(0.0299)	(0.0388)
One Year Before Adoption of Castle Doctrine Law				-0.0212		
				(0.0246)		
Observations	550	550	550	550	550	550
Panel B: Log Homicide Rate (OLS - Unweighted)						
Castle Doctrine Law	0.0877	0.0811	0.0600	0.0461	0.0580	0.0672
	(0.0638)	(0.0769)	(0.0684)	(0.0764)	(0.0662)	(0.0450)
One Year Before Adoption of Castle Doctrine Law				-0.0557		
				(0.0494)		
Observations	550	550	550	550	550	550
State and Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Region-by-Year Fixed Effects		Yes	Yes	Yes	Yes	Yes
Time-Varying Controls			Yes	Yes	Yes	Yes
Contemporaneous Crime Rates					Yes	
State-Specific Linear Time Trends						Yes

Conclusions

These laws do lead to an 8% net increase in homicide rates, translating to around 600 additional homicides per year across the 21 adopting states

Unlikely that all of the additional homicides were legally justified

Economics of crime and behavior: incentives matter in some contexts

Table of Contents